# Data Literacy: Making Better Decisions with Evidence

## What data literacy means

Data literacy is the ability to read, question, interpret, and communicate information from data. It does not require everyone to become a statistician. It does require understanding where numbers came from, what they measure, how they may be incomplete, and what conclusions they can reasonably support.

Data appears in everyday decisions: comparing prices, evaluating a health claim, tracking a budget, or reviewing a business dashboard. A number can look precise while still being misleading if its definition or context is unclear. Data literacy encourages people to ask what was measured, who was included, when it was collected, and what may be missing.

## Questions to ask about a dataset

Start with the source. Was the information collected directly, reported by an organization, or copied from another source? Check the time period, population, and units. A dataset about monthly sales is not directly comparable with one about annual revenue unless the time basis is aligned.

Look for missing values, duplicate records, unusual values, and inconsistent categories. Missing information may not be random: for example, people with limited internet access may be underrepresented in an online survey. Data cleaning can improve consistency, but decisions made during cleaning should be documented because they can affect results.

## Summaries and visualizations

A summary statistic compresses information into a few numbers. The mean is sensitive to extreme values, while the median often better describes a typical observation in a skewed distribution. A range or interquartile range can communicate spread. No single statistic tells the whole story, so summaries should be chosen to fit the question.

Charts help reveal patterns, but design matters. Bar charts are useful for comparing categories; line charts often show change over time; scatterplots can reveal relationships between two numerical variables. Axes should be labeled, units should be clear, and scales should not exaggerate differences. A chart should make the important comparison easy to see without hiding relevant context.

## Correlation, causation, and uncertainty

Two variables may move together without one causing the other. A third factor may influence both, or the relationship may be coincidental. Establishing causation usually requires stronger evidence than observing a correlation, such as a well-designed experiment or a credible causal analysis.

Uncertainty is a normal part of measurement and inference. A survey estimate may vary because a different sample would produce a different result. A forecast depends on assumptions and can become less reliable farther into the future. Communicating uncertainty honestly is not a weakness; it helps decision-makers understand how much confidence to place in a conclusion.

## Fairness, privacy, and responsible use

Data can reflect existing inequalities. If a system is trained on incomplete or biased records, its outputs may reproduce those patterns. Analysts should examine whether groups are represented fairly and whether errors affect some people more than others. A technically accurate model can still be inappropriate if the data or decision process is unfair.

Privacy matters when data contains information about identifiable people. Collect only what is needed, limit access, and remove or protect sensitive details where possible. Aggregated statistics can still reveal information when groups are very small. Responsible data work considers not only what can be measured, but also whether it should be collected and how the results may be used.

## Communicating findings

A useful data explanation connects a question, evidence, interpretation, and limitation. State the main result in plain language, then show the measure or chart that supports it. Explain what the result does not establish, and avoid implying certainty that the data cannot provide.

Before sharing an analysis, check that figures are reproducible, labels are understandable, and conclusions match the evidence. Invite questions and be willing to revise a claim when new information appears. Data literacy is ultimately a habit of careful reasoning: use evidence, make assumptions visible, and remain open to correction.